higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

MATHEMATICS
(First Paper)
NQF LEVEL 4

(10401064)

21 February 2019 (Y-Paper)
13:00–16:00

Nonprogrammable calculators may be used.

This question paper consists of 9 pages, a formula sheet of 2 pages and 3 answer sheets.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Use a BLACK or BLUE pen.
5. Leave at least THREE lines after each question.
6. Start each section on a NEW page.
7. Diagrams are NOT drawn to scale.
8. Where necessary, answers should be rounded off to THREE decimal places, unless stated otherwise.
9. Write neatly and legibly.

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QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the correct option and write only the letter (A, B, C or D) next to the question numbers (1.1.1–1.1.5) in your ANSWER BOOK. [**Note:** $\boldsymbol{i = \sqrt{-1}}$]

1.1.1 If $f(x) = x^2 + 25$ then $f(x) = 0$ if …

A $x = 625$
B $x = 5$ or $x = -5$
C $x = -5i$ or $x = 5i$
D None of the above

1.1.2 Simplify $(7 - 3i)(7 + 3i)$.

A 58
B $49 - 42i - 9i^2$
C 40
D $49 - 9i$

1.1.3 Simplify $(2 - 7i)^2$.

A $-45 + 28i$
B $4 + 49i^2$
C $2 - 28i$
D $-45 - 28i$

1.1.4 If $z = 7 + 4i$, determine $\bar{z} - z$.

A $-8i$
B 0
C $-14 - 8i$
D -65

1.1.5 Solve for *x* and *y*, given that $-2 - 4yi = 4x + 16i$.

A $x = 2$ and $y = 12$
B $x = 4$ and $y = 2$
C $x = \frac{1}{2}$ and $y = 4$
D $x = -\frac{1}{2}$ and $y = -4$

(5 × 1) (5)

1.2 Simplify the following and leave the answer in rectangular form. Show all working.

$$\frac{(2-3i)-(3+2i)}{(3-2i)-(2-i)}$$ (3)

1.3 The Argand diagram in FIGURE 1 below represents two complex numbers in the polar form:

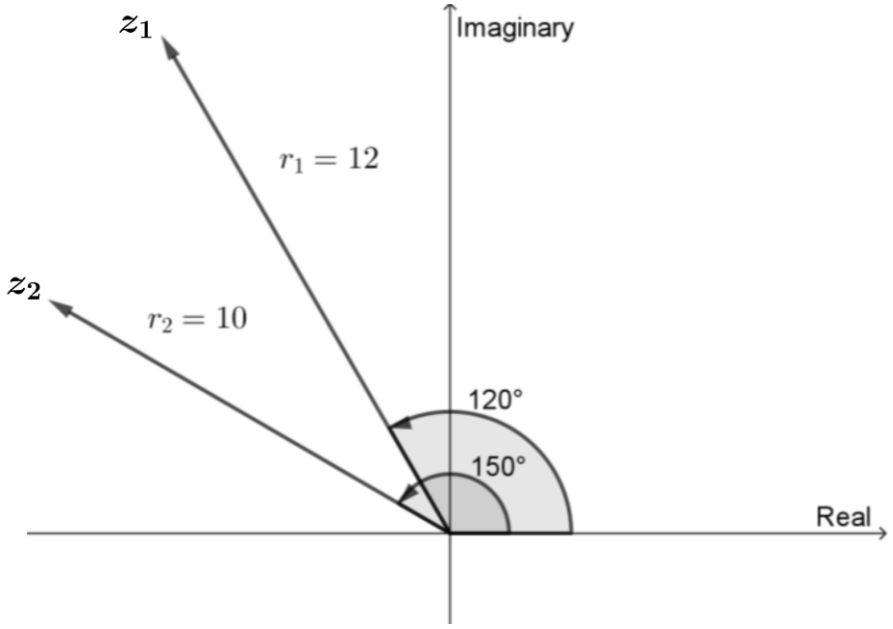


FIGURE 1

1.3.1 Use the information in the Argand diagram in FIGURE 1 to express z_1 and z_2 in polar form (1)

1.3.2 Hence, calculate $z_1 + z_2$, showing all working. Give your answers in rectangular form, correct to two decimal places. (3)

1.4 Apply De Moivre's theorem to calculate the following. Leave your answer in polar form with a positive argument.

$$\frac{\sqrt{18}\left(cis\,30°\right)^3}{\left(\sqrt{2}\,cis\,45°\right)^4}$$ (3)

1.5 Solve for x and y if: $(4+y)i^5 - (8x+y+4)i^2 = 4(x^2+3i)$. (5)

[20]

QUESTION 2

2.1 Use differentiation rules to determine the derivatives of the following, leaving your answers with POSITIVE EXPONENTS and in SURD form where applicable:

2.1.1 $f(x) = 4x^3 - \frac{5}{\sqrt[3]{x^2}} - \frac{a}{x} - \frac{\ln x}{4} - 3^4$ where a is a constant (5)

2.1.2 $y = \frac{e^{4x}}{\tan 4x}$ (3)

2.1.3 $y = \left(3x^2 - 5\right)^5$ (3)

2.2 The function $s = t^3 \ln t$ gives the distance of an object from a point, where s is distance in metres and t is the time in seconds.

2.2.1 Determine the distance when t = 2, correct to 3 decimal places. (1)

2.2.2 Determine the equation for the acceleration of the object. (3)

2.3 Determine the derivative of $f(x) = 3 - 2x$ using first principles. (2)

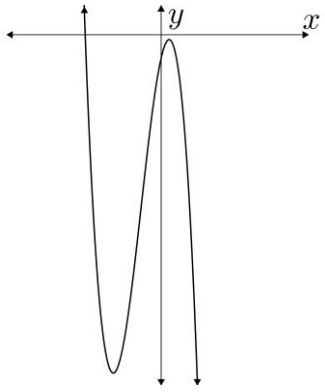
2.4 Given: $f(x) = 4x^3 + 15x^2 - 18x + 6$.

2.4.1 Determine the x coordinates of the turning points of f. (3)

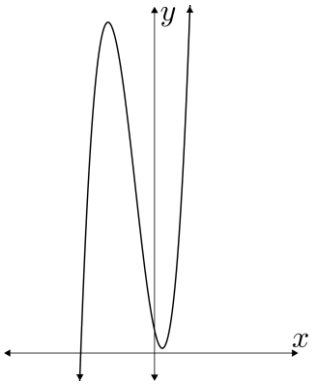
2.4.2 By applying $\frac{d^2y}{dx^2}$, show which x value produces a local maximum and which x value produces a local minimum. (2)

2.4.3 Determine the coordinates of the point of inflection of f. (2)

2.4.4 Which graph below best represents the function f, graph 1 or graph 2?



GRAPH 1



GRAPH 2

(1)

[25]

QUESTION 3

3.1 Determine the following integrals and simplify your answers. Leave your answers with POSITIVE exponents and in SURD form where applicable:

3.1.1 $\int (\frac{9}{4x} - \frac{12}{\sqrt{4x}} + e^{\frac{x}{2}} + 4\sec^2 4x)\,dx$ (5)

3.1.2 $\int (2x + \sin 3x + 4x^3 + e^{3x})dx$ (2)

3.2 The velocity equation of an object is given by $v = t^2 - 5t + 12$ where v is the velocity in metres per second and t is the time in seconds.

Determine the distance travelled between t = 0 and t = 3. (3)

3.3 Refer to the sketch below. A game reserve has a river (broken line) as one of its boundaries. A group of level-4 students used several points on a map of the river to arrive at the sample regression equation of the form $f(x) = x^3 - 6x^2 + 9x + 2$ which best represents the river boundary.

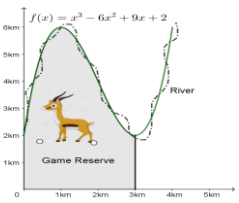


FIGURE 2

Calculate the magnitude of the shaded area by making use of integration. The game reserve is bounded by the graph of f, the x axis, the line $x = 0$, and the line $x = 3$. (5)

[15]

QUESTION 4

4.1 If the function $g(x) = 2x^3 - x^2 - px + 20$ is divided by $x + 4$, the remainder is 4. Calculate the value of p by applying the remainder theorem. (2)

4.2 $2x - 1$ is a factor of the polynomial $f(x) = 2x^3 - x^2 - ax + b$, and the point $(2;-36)$ lies on the graph of f. Determine the values of a and b. (5)

4.3 FIGURE 3 below shows the graphs of the functions g^{-1} and h where:

$g(x) = a^x$ and

$h(x) = mx + c$ over the interval $-1 \leq x < 6, x$ a real number.

The graph of h cuts the x axis at (3; 0) and the y axis at (0; -2).

The point C(3;1) lies on the graph of g^{-1}.

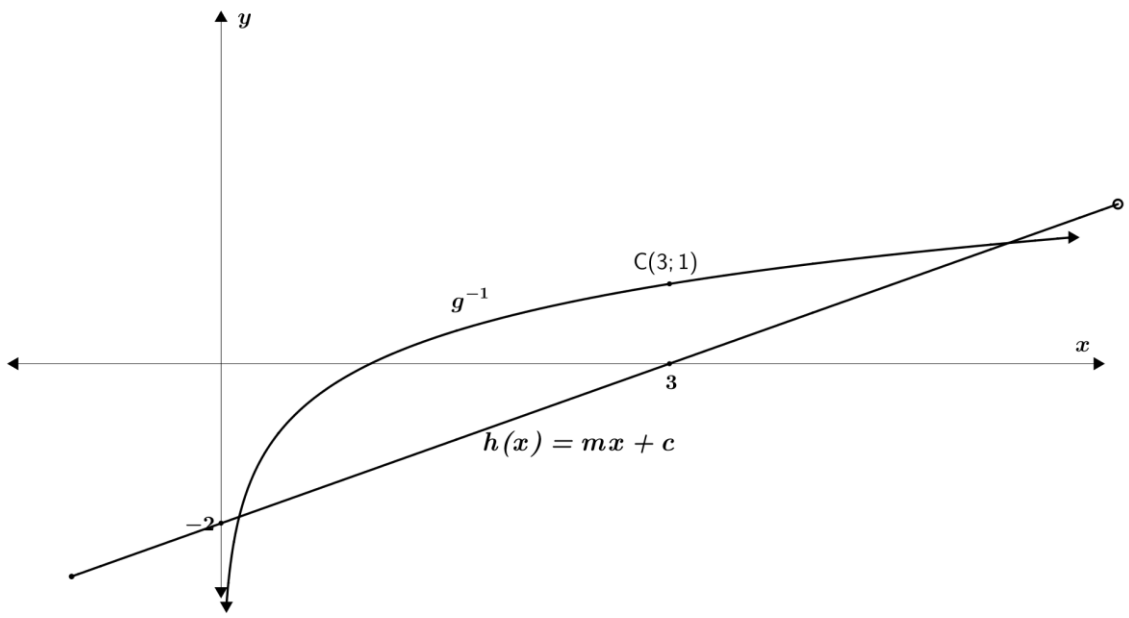


FIGURE 3

4.3.1 Determine the values of m and c. (2)

4.3.2 Determine the range of $h(x)$. (2)

4.3.3 Use the information given about g^{-1} to determine the equation of g. (1)

4.3.4 Hence write down the equation of g^{-1} in the form $x = \cdots$. (1)

4.3.5 The graph of g^{-1} is drawn on ANSWER SHEET 1. Draw a neat graph of g on the same system of axes, clearly showing at least two points on g. (3)

4.3.6 Give the equation of the line about which the graphs of g and g^{-1} are symmetrical (1)

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4.4 Given: $f(x) = x^3 - 12x - 16$. The x coordinates of the turning points are given as $x = 2$ and $x = -2$.

4.4.1 Determine the x intercept(s) and the y intercepts of $f(x) = x^3 - 12x - 16$. Show ALL calculations. (5)

4.4.2 Hence, draw a neat graph of $f(x) = x^3 - 12x - 16$, clearly showing the intercepts with the axes and the coordinates of the turning points. Use ANSWER SHEET 2. (3)

[25]

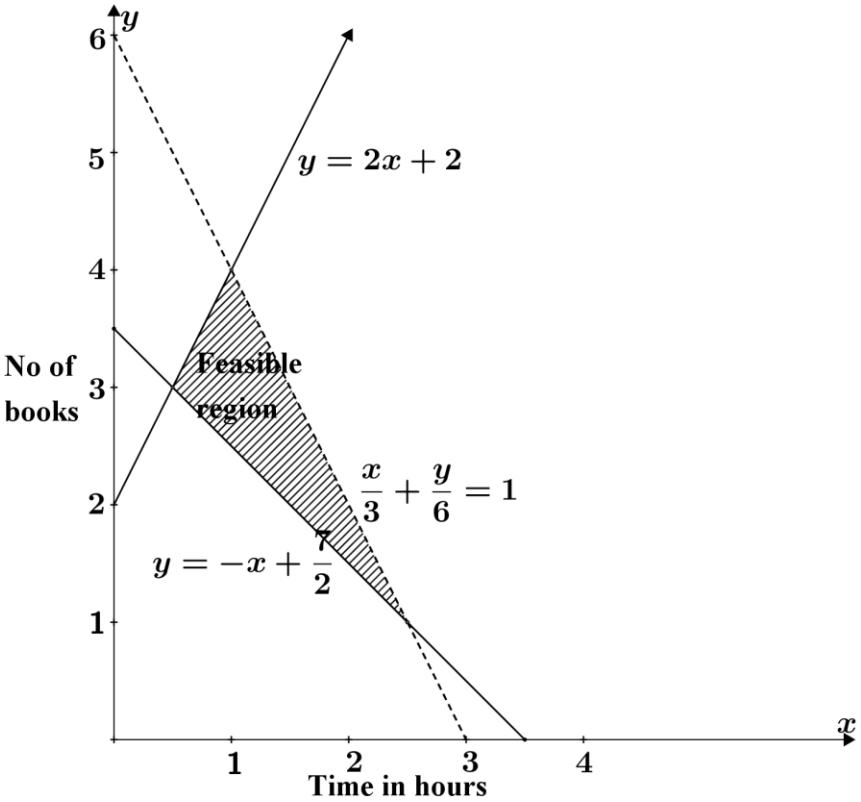
QUESTION 5

5.1 The following diagram models a linear programming problem in which the x axis shows the time in hours and the y axis indicates the number of books that can be read in x hours . The feasible region is bounded by three straight lines, one of which is a dotted line. The equations of the straight lines are given by:

$y = 2x + 2$,

$y = -x + \frac{7}{2}$ and

$\frac{x}{3} + \frac{y}{6} = 1$



5.1.1 Write down the constraints that describe the feasible region. (3)

5.1.2 Is it feasible to read exactly two books in two hours?
Motivate your answer using the given model. (1)

5.2 Vincent Cele from the Western Cape makes boerewors by mixing two types of mincemeat, beef (x) and pork (y). He requires, in total, at least 9 kg of mincemeat, but not more than 15 kg, to make the boerewors. The boerewors must contain at least 2 kg of beef for every 1 kg of pork. Vincent must use at least 7 kg of beef.

5.2.1 Two of the constraints are $x \geq 0$ and $y \geq 0$.
Model the other constraints in respect of the above information in terms of x and y. (3)

5.2.2 The beef cost R80 per kg and the pork cost R60 per kg.
Write the equation that represents the total cost (C) in terms of x and y. (1)

5.2.3 Make a graphical representation of the constraints in 5.2.1 on ANSWER SHEET 3. Shade in the feasible region. (5)

5.2.4 Use a search line to determine the x and y values for the most costly mix. (2)
[15]

TOTAL: 100

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FORMULA SHEET

1. $z = rcos\theta \pm rj\sin\theta$ or $z = rcos\theta \pm risin\theta$

2. $z = a \pm bj \ or \ z = a \pm bi \qquad where\ i = j = \sqrt{-1}$

3. $r\angle\theta = r \angle \theta = rcis\ \theta$ where $'r'$ represent modulus and $'\theta'$ represent argument

4. $m = \lim\limits_{h\to 0} \dfrac{f(x+h) - f(x)}{h}$

5. $\dfrac{d}{dx} x^n = nx^{n-1}$

6. $\dfrac{d}{dx} k = 0$

7. $\dfrac{dy}{dx} = u.\dfrac{dv}{dx} + v.\dfrac{du}{dx}$ **OR** $\dfrac{d}{dx}\left[f(x).g(x)\right] = f(x)\ g'(x) + f'(x)\ g(x)$

8. $\dfrac{dy}{dx} = \dfrac{v\dfrac{du}{dx} - u\dfrac{dv}{dx}}{v^2}$ **OR** $\dfrac{d}{dx}\left[\dfrac{f(x)}{g(x)}\right] = \dfrac{g(x)\ f'(x) - f(x)\ g'(x)}{[g(x)]^2}$

9. $\dfrac{dy}{dx} = \dfrac{du}{dx} \times \dfrac{dy}{du}$ **OR** $\dfrac{d}{dx} f\left(g(x)\right) = f'\left(g(x)\right) g'(x)$

10. If $y = (f(x))^n$ then $\dfrac{dy}{dx} = n(f(x))^{n-1}.\ f'(x)$

11. If $y = \ln kx$ then $\dfrac{dy}{dx} = \dfrac{k}{kx}$ **OR** If $f(x) = \ln kx$ then $f'(x) = \dfrac{k}{kx}$

12. If $y = e^x$ then $\dfrac{dy}{dx} = e^x$ **OR** If $f(x) = e^x$ then $f'(x) = e^x$

13. If $y = e^{kx}$ then $\dfrac{dy}{dx} = ke^{kx}$ **OR** If $f(x) = e^{kx}$ then $f'(x) = ke^{kx}$

14. If $y = \sin x$ then $\dfrac{dy}{dx} = \cos x$ **OR** If $f(x) = \sin x$ then $f'(x) = \cos x$

15. If $y = \cos x$ then $\dfrac{dy}{dx} = -\sin x$ **OR** If $f(x) = \cos x$ then $f'(x) = -\sin x$

16. If $y = \tan x$ then $\dfrac{dy}{dx} = \sec^2 x$ **OR** If $f(x) = \tan x$ then $f'(x) = \sec^2 x$

17. If $y = \cot x$ then $\dfrac{dy}{dx} = -\cos ec^2 x$ **OR** $f(x) = \cot x$ then $f'(x) = -\cos ec^2 x$

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18. If $y=\sec x$ then $\frac{dy}{dx}=\sec x\tan x$ or **OR** $f(x)=\sec x$ then $f'(x)=\sec x\tan x$

19. If $y=\cos ecx$ then $\frac{dy}{dx}=-\cos ecx\cot x$ **OR** $f(x)=\cos ecx$ then $f'(x)=-\cos ecx\cot x$

20. $\int x^n\,dx = \frac{x^{n+1}}{n+1}+c$

21. $\int k\,x^n\,dx = k\int x^n\,dx$ where k is a constant value

22. $\int[f(x)\pm g(x)]dx=\int f(x)\,dx\pm\int g(x)\,dx$

23. $\int\frac{k}{x}\,dx = k\ln x+c$

24. $\int e^{kx}\,dx=\frac{e^{kx}}{k}+c$

25. $\int\sin kx\,dx = \frac{-\cos kx}{k}+c = -\frac{1}{k}\cos kx+c$

26. $\int\cos kx\,dx = \frac{\sin kx}{k}+c = \frac{1}{k}\sin kx+c$

27. $\int\sec^2 kx\,dx = \frac{1}{k}\tan kx+c$

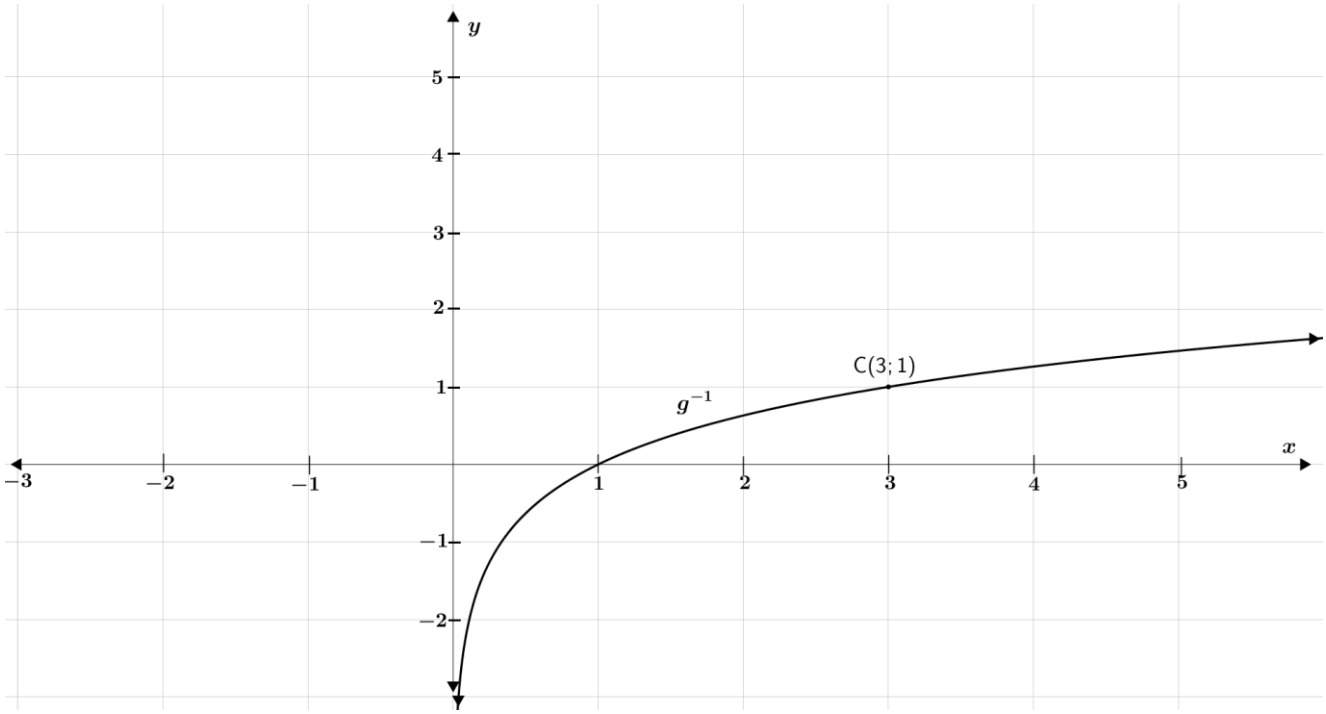
28. $A=\int_a^b ydx$

29. $\left[f(x)\right]_a^b = f(b)-f(a)$

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ANSWER SHEET 1 EXAMINATION NUMBER:

QUESTION 4.3.5



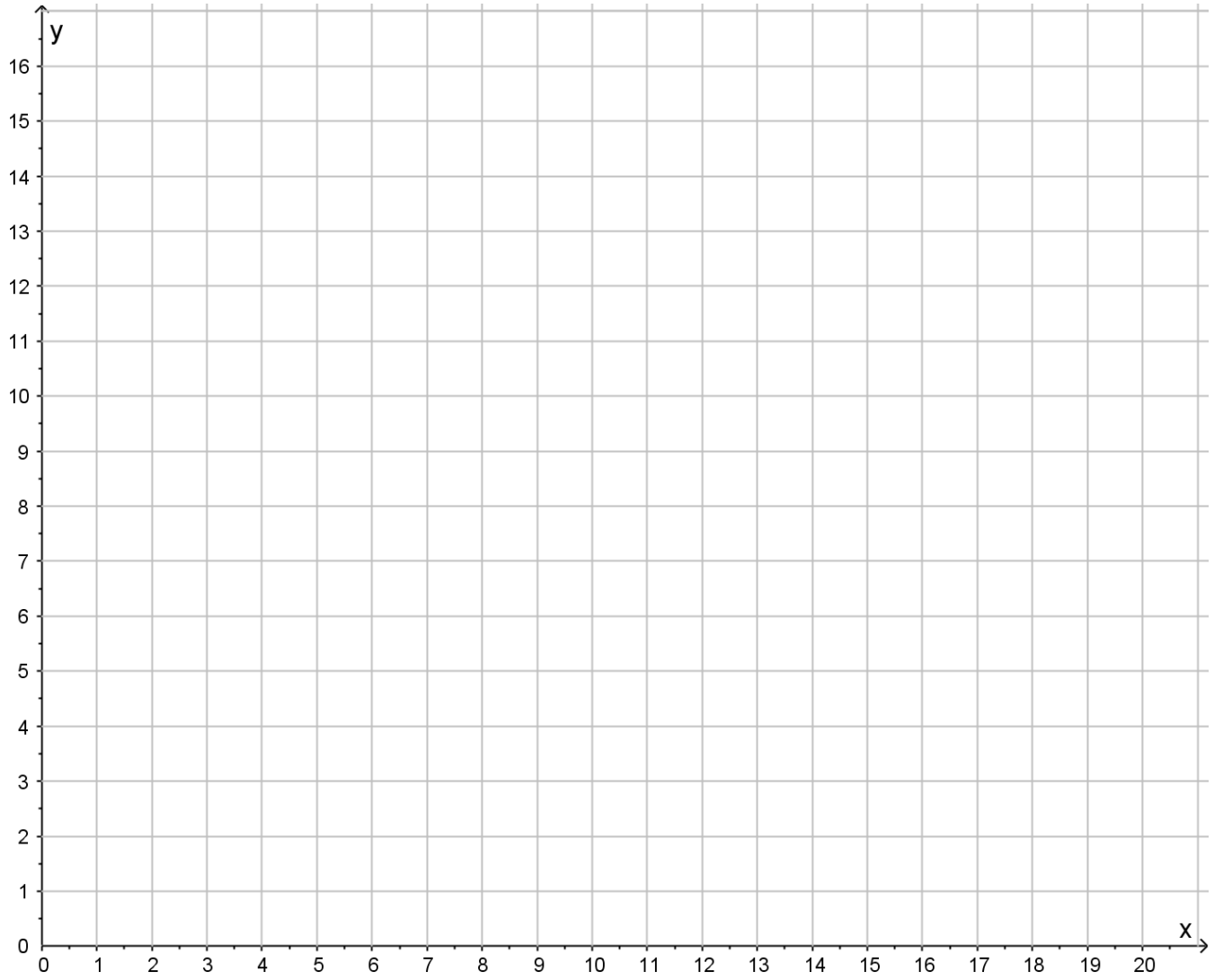
ANSWER SHEET 2 EXAMINATION NUMBER:

QUESTION 4.4.2

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ANSWER SHEET 3 EXAMINATION NUMBER:

QUESTION



5.2.3

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